

Rushikesh Dhumal

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Education

Master of Science in Data Science, Rutgers University, New Brunswick
Microsoft Certified: Azure Data Scientist Associate DP-100

August 2024 — May 2026
July 2025

Skills

Programming & Tools: Python, R, SQL, Pandas, NumPy, PySpark, Git, Docker

Machine Learning & AI: Scikit-learn, TensorFlow, PyTorch, MLflow, SHAP, Ensembles, LangChain, RAG

Cloud & Databases: PostgreSQL, Snowflake, Azure, Spark, Databricks, dbt, Airflow

Visualization & BI: Tableau, Power BI, Superset, Looker Studio, Microsoft Excel

Experience

AI/ML Software Engineering Intern | Micronotes

June 2025 – December 2025

- Accelerated predictive analytics delivery for 50+ clients by 2x by building a PySpark ETL pipeline to ingest and wrangle 230M+ Experian credit records into a structured data warehouse.
- Reduced data integrity errors by 30% across large-scale heterogeneous datasets by deploying statistical anomaly detection using Z-score and IQR-based pattern recognition.
- Provided consultative services to 5 loan channel teams by automating analytical reporting workflows, reducing manual data processing time by 40%.

Data Scientist | Fields Data

July 2022 – July 2024

- Improved pattern recognition accuracy by 70% for 600+ organizations by building an NLP engine using spaCy to extract insights from unstructured organizational and project profile data.
- Prioritized resource allocation across 120+ at-risk regions by applying K-Means clustering and data mining techniques on large heterogeneous disaster-preparedness datasets.
- Provided consultative services to business stakeholders by translating data analysis findings into actionable recommendations, driving 80% efficiency gain in cross-functional reporting.
- Reduced reporting latency by 40% for BI dashboards by optimizing SQL joins, indexing strategies, and CTEs on AWS RDS.

Data Scientist Intern | Nestle

March 2022 – May 2022

- Quantified distribution effectiveness with 95% statistical significance by applying probability theory and hypothesis testing in A/B tests across 1,600+ retail outlets.
- Drove 15% improvement in resource allocation decisions by leading discovery sessions with stakeholders to identify business requirements and delivering data-driven insights through Tableau dashboards.

Data Scientist Intern | Fields Data

April 2021 – May 2022

- Expanded data processing capacity by 25% by refining ETL pipelines ingesting from 5 sources, including Oracle NetSuite.
- Identified top 8 key predictors for disaster response prioritization models by performing EDA with correlation analysis, variance inflation factor testing, and feature importance ranking on disaster-impact variables.
- Reduced data quality errors by 25% by building validation checks and implementing data governance controls, ensuring data integrity and accuracy across ingested datasets.

Projects

PyTorch LLM Pretraining System: 124M to 1.3B GPT Multi-Node Scaling on HPC (DDP/FSDP)

- Built decoder-only GPT models (124M -> 1.3B) from scratch in pure PyTorch on HPC: single-GPU -> DDP -> multi-node FSDP, with AMP, cosine LR, grad clipping, activation checkpointing, and DCP sharded checkpointing on SLURM.
- Designed an eval & benchmarking harness (val perplexity, GPT-2 baselines at matched scales, FSDP to single-GPU export) and debugged distributed failures (NCCL rendezvous, OOM, scratch quota) through log-driven root-cause analysis.

F1 Telemetry Strategy Prediction | Hybrid GRU Deep Learning Model

- Engineered 38 leakage-safe features from 440K multi-season F1 telemetry laps, modeling lap-stint-race temporal hierarchy with race-season GroupKFold validation to prevent inter-lap leakage and ensure production-equivalent evaluation.
- Developed a dual-branch Hybrid GRU+FC architecture with learned driver/circuit embeddings, combining sequential tire-wear trajectories with race-strategy features; outperformed a tuned LightGBM baseline by +3.5pp.
- Analyzed ensemble boost-decay and inter-model correlation across 10 neural architectures, identifying structural diversity limits that informed targeted Optuna hyperparameter optimization and improved model selection efficiency.

Urban Mobility Analytics Platform (Snowflake + dbt + Airflow + Superset)

- Validated demand forecasting hypotheses on 94M+ NYC TLC public records by building a time-series ML model with lag and rolling features, outperforming the naive baseline on MAPE and tracking experiments with MLflow for reproducibility.
- Engineered an end-to-end ELT pipeline on Snowflake and dbt with automated Airflow orchestration, surfacing demand heatmaps and real-time KPIs in Superset to translate large-scale public datasets into analytical insights.